

Olivine Silier

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pronunciation: Aw-luh-veen See-lee-ay



Research Program

I am broadly interested in **harmonic analysis** and **discrete geometry**. I am currently working on **incidence geometry** in the Euclidean plane, as well as **multidimensional Van der Corput** type bounds.

Education

- 2022–2027 **PhD**, *University of California Berkeley*, Mathematics.
advisor: Prof. Ruixiang Zhang.
- 2018–2022 **Bachelor of Science**, *California Institute of Technology*, Mathematics.
advisor: Prof. Nets Katz.

Publications

- 2026 Yuda Chen, Olivine Silier and Ruixiang Zhang. *Improved multidimensional sublevel set estimates* in preparation.
- 2026 Teddy Astor, Nets Katz and Olivine Silier. *Spacing properties of extremal Szemerédi-Trotter examples*. in preparation.
- 2026 Nets Katz and Olivine Silier. *Structure of cell decompositions in extremal Szemerédi-Trotter examples*. arXiv:2303.17186.
- 2025 Larry Guth and Olivine Silier. *Sharp Szemerédi-Trotter constructions in the plane*. *Electronic Journal of Combinatorics*; 32 (1).
- 2023 Adam Sheffer and Olivine Silier. *A Structural Szemerédi-Trotter Theorem for Cartesian Products*. *Discrete & Computational Geometry*; 646–666.

Awards and Fellowships

- 2025–2026 **Simons Foundation**, *Dissertation Fellowship in Mathematics*, \$30,000 in summer support
- 2025–2026 **UC Berkeley**, *Berkeley Connect Fellowship*, competitive graduate student instructorship
- 2025 **UC Berkeley Mathematics Department**, *Outstanding Graduate Student Instructor*
- 2023/24 **Jane Street**, *Graduate Research Fellowship*, one year tuition + stipend.
- 2022/23 **Egon & Joan von Kaschnitz**, *IHouse Gateway Fellowship*, room and board + \$5,000.
- 2022 **Berkeley University**, *Graduate Fellowship*, one semester and summer support.
- 2022 **Caltech**, *Eric Temple Bell Undergraduate Mathematics Research*, cash prize.